

D326

Advanced Data Management

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A retail chain with 250 stores has a table OrderLine(order_id, product_id, quantity, product_name) with primary key (order_id, product_id). The table was built in PostgreSQL 15. Which normal form does it violate, and why?

- A. 1NF, because the table has four columns.
- B. None; any table with a primary key is normalized.
- C. 3NF, because product_name depends on another non-key column.
- D. 2NF, because product_name depends on only part of the composite key.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A table Employee(emp_id, dept_id, dept_name) has primary key emp_id. dept_name is determined by dept_id. Which normal form does it violate?

- A. 3NF
- B. 2NF
- C. It is fully normalized.
- D. 1NF

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An analyst at a sporting goods company, which sponsors a marathon, designs a star schema for sales. Columns include sale_amount, units_sold, product_category, store_city, and date_key. The warehouse runs on a cloud platform. Which columns belong in the fact table?

- A. All five columns
- B. sale_amount, units_sold, and date_key along with other dimension keys
- C. product_category and store_city
- D. Only date_key

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A streaming company with 9 million subscribers wants to store raw clickstream logs, images, and JSON event files in their original formats for future data science work that is not yet defined. The storage budget was approved in June. What should it build?

- A. A data warehouse
- B. A data lake
- C. An OLTP database
- D. A data mart

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A customers table has 10 rows. An orders table has 15 rows, and those orders belong to 7 different customers. The database server has 64 GB of RAM. How many rows does `SELECT * FROM customers c LEFT JOIN orders o ON c.id = o.customer_id` return?

- A. 150
- B. 10
- C. 18
- D. 15

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Using the same tables (10 customers; 15 orders from 7 customers), how many rows does a LEFT JOIN from customers to orders return when filtered `WHERE o.customer_id IS NULL`?

- A. 8
- B. 7
- C. 0
- D. 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An analyst at a hotel group with 60 properties wants each property whose total 2025 revenue exceeded \$5 million. The query groups by property_id. The data comes from the company's reservation system. Which clause filters on the total?

- A. WHERE revenue > 5000000
- B. HAVING SUM(revenue) > 5000000
- C. ORDER BY SUM(revenue) DESC
- D. WHERE SUM(revenue) > 5000000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A query must count orders per region, including only orders placed in 2025. Where does the 2025 condition go?

- A. In the WHERE clause
- B. In the GROUP BY clause
- C. In the SELECT list
- D. In the HAVING clause

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A payroll database at a company with 4,000 employees must copy the old salary into an audit table automatically every time a salary is updated, no matter which app makes the change. The DBA prefers PL/pgSQL. What should be created?

- A. A view of the salary table
- B. An index on the salary column
- C. An UPDATE trigger on the salary table
- D. A stored procedure that apps call after updates

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A customer who has bought from a furniture retailer since 2012 moves from the West region to the East region. Sales reports must keep her past purchases credited to West. The warehouse refreshes nightly. How should the customer dimension be updated?

- A. Type 1: overwrite the region with East.
- B. Delete the customer and re-add her.
- C. Type 2: add a new row with the East region and mark the old row as expired.
- D. Update the fact table's region column.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A customer's misspelled last name must be corrected, and no one needs the old spelling. Which SCD type fits?

- A. Type 1
- B. Add a column for every past spelling
- C. Type 2
- D. Rebuild the whole warehouse

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A query on a 30-million-row shipments table, which the logistics team runs 500 times a day, filters on tracking_number and takes 12 seconds. The table was migrated from an older system. Each tracking number is unique. What is the best fix?

- A. Remove the primary key to speed up reads.
- B. Run the query 500 fewer times.
- C. Index every column in the table.
- D. Create an index on tracking_number.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A retail chain ~~with 250 stores~~ has a table OrderLine(order_id, product_id, quantity, product_name) with primary key (order_id, product_id). ~~The table was built in PostgreSQL 15.~~ Which normal form does it violate, and why?

Target word: identify the normal-form violation **Decides:** product_name depends only on product_id, part of the composite key.

A	Wrong stem number	Column count does not decide 1NF; the values are atomic.
B	Absolute language	A key alone does not remove partial dependencies.
C	Confusion set	product_id is part of the key, so this is partial, not transitive.
D	Correct	A partial dependency breaks 2NF.

2. Answer A Concept 1 · transfer

A table Employee(emp_id, dept_id, dept_name) has primary key emp_id. dept_name is determined by dept_id. Which normal form does it violate?

Target word: identify a transitive dependency **Decides:** A non-key column depends on another non-key column.

A	Correct	emp_id → dept_id → dept_name is transitive.
B	Inverted operation	2NF concerns composite keys; this key is a single column.
C	Dropped element	Ignores the dept_id to dept_name dependency.
D	Adjacent category	Values are atomic.

3. Answer B Concept 2

An analyst at a sporting goods company, ~~which sponsors a marathon,~~ designs a star schema for sales. Columns include sale_amount, units_sold, product_category, store_city, and date_key. ~~The warehouse runs on a cloud platform.~~ Which columns belong in the fact table?

Target word: fact table contents **Decides:** Measures plus foreign keys belong in the fact table.

A	Absolute language	Putting text attributes in the fact table defeats the star design.
B	Correct	Numeric measures and keys to dimensions.
C	Inverted operation	Descriptive attributes belong in dimensions.
D	Dropped element	Leaves out the measures being analyzed.

4. Answer B Concept 3

A streaming company ~~with 9 million subscribers~~ wants to store raw clickstream logs, images, and JSON event files in their original formats for future data science work that is not yet defined. ~~The storage budget was approved in June.~~ What should it build?

Target word: storage architecture **Decides:** Raw, mixed-format data with no defined schema yet.

A	Adjacent category	Requires cleaned, structured data modeled up front.
B	Correct	Stores raw data of any type; schema is applied on read.
C	Inverted operation	Built for transactions, not storing raw files for analysis.
D	Confusion set	A subject-focused subset for one department.

5. Answer C Concept 4

A customers table has 10 rows. An orders table has 15 rows, and those orders belong to 7 different customers. The database server has 64 GB of RAM. How many rows does `SELECT * FROM customers c LEFT JOIN orders o ON c.id = o.customer_id` return?

Target word: LEFT JOIN row count **Decides:** All 15 matched rows plus one row for each of the 3 customers with no orders.

A	Adjacent category	10 × 15 is a CROSS JOIN.
B	Dropped element	Ignores that customers with several orders appear several times.
C	Correct	15 matched rows + 3 unmatched customers with NULL order columns.
D	Confusion set	That is the INNER JOIN count.

6. Answer D Concept 4 · transfer

Using the same tables (10 customers; 15 orders from 7 customers), how many rows does a LEFT JOIN from customers to orders return when filtered `WHERE o.customer_id IS NULL`?

Target word: customers with no orders **Decides:** Only unmatched left rows have NULL order columns.

A	Wrong stem number	15 – 7 mixes order and customer counts.
B	Inverted operation	Counts customers who do have orders.
C	Confusion set	Applies INNER JOIN logic, which drops unmatched rows.
D	Correct	10 – 7 = 3 customers with no orders.

7. Answer B Concept 5

An analyst at a hotel group with 60 properties wants each property whose total 2025 revenue exceeded \$5 million. The query groups by `property_id`. The data comes from the company's reservation system. Which clause filters on the total?

Target word: filter on an aggregate **Decides:** The condition uses SUM after grouping.

A	Dropped element	Filters single rows, not totals.
B	Correct	HAVING filters groups after aggregation.
C	Adjacent category	Sorts but does not filter.
D	Confusion set	WHERE cannot use aggregate functions.

8. Answer A Concept 5 · transfer

A query must count orders per region, including only orders placed in 2025. Where does the 2025 condition go?

Target word: filter rows before grouping **Decides:** The condition is on individual rows.

A	Correct	Row filters apply before GROUP BY.
B	Adjacent category	GROUP BY defines groups; it does not filter.
C	Confusion set	SELECT chooses columns, not rows.
D	Inverted operation	HAVING is for conditions on groups.

9. Answer C Concept 6

A payroll database ~~at a company with 4,000 employees~~ must copy the old salary into an audit table automatically every time a salary is updated, no matter which app makes the change. ~~The DBA prefers PL/pgSQL.~~ What should be created?

Target word: object that fires on data change **Decides:** Automatically on every update, regardless of app.

A	Adjacent category	A view presents data; it does not act on changes.
B	Confusion set	Speeds lookups; records nothing.
C	Correct	Triggers fire automatically on table events.
D	Dropped element	Depends on every app remembering to call it.

10. Answer C Concept 7

A customer who has bought from a furniture retailer ~~since 2012~~ moves from the West region to the East region. Sales reports must keep her past purchases credited to West. ~~The warehouse refreshes nightly.~~ How should the customer dimension be updated?

Target word: SCD type that keeps history **Decides:** Past facts must stay tied to the old region.

A	Inverted operation	Rewrites history; past sales move to East.
B	Absolute language	Breaks links to her past facts.
C	Correct	Keeps history so past sales stay with West.
D	Adjacent category	Region is a dimension attribute.

11. Answer A Concept 7 • transfer

A customer's misspelled last name must be corrected, and no one needs the old spelling. Which SCD type fits?

Target word: SCD type when history is not needed **Decides:** A correction with no need for history.

A	Correct	Overwrite; history of a typo has no value.
B	Adjacent category	Type 3-style tracking is unneeded here.
C	Inverted operation	Keeps history that no one needs.
D	Absolute language	A simple update is enough.

12. Answer D Concept 8

A query on a 30-million-row shipments table, which the logistics team runs 500 times a day, filters on tracking_number and takes 12 seconds. ~~The table was migrated from an older system.~~ Each tracking number is unique. What is the best fix?

Target word: index choice **Decides:** A highly selective column in a frequent WHERE clause.

A	Inverted operation	Removes an index, making reads slower.
B	Wrong stem number	Usage count does not fix the slow lookup.
C	Absolute language	Slows writes and wastes storage without helping this query more.
D	Correct	Unique values make the index highly selective.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Normal forms	1NF has atomic values and no repeating groups; 2NF removes partial dependencies on part of a composite key; 3NF removes transitive dependencies between non-key columns.	Calling a transitive dependency a 2NF problem, or thinking 2NF matters with a single-column key.	A non-key column that depends on only part of the key, or on another non-key column.
2	Fact versus dimension tables	In a star schema, the fact table holds numeric measures and foreign keys at a stated grain; dimension tables hold descriptive attributes used to filter and group.	Putting descriptive text like product category in the fact table.	Whether a column is something you sum or something you group by.
3	Warehouse, lake and mart	A data warehouse stores cleaned, structured, integrated data (schema-on-write); a data lake stores raw data of any type in native format (schema-on-read); a data mart is a subject-focused subset for one department.	Calling a department's small subset a data lake, or loading raw logs into a warehouse unchanged.	Raw versus modeled data, and whole enterprise versus one department.
4	INNER versus LEFT JOIN	INNER JOIN returns only matching rows; LEFT JOIN returns every left-table row, with NULLs where no match exists.	Expecting a LEFT JOIN to drop unmatched rows, or an INNER JOIN to keep them.	Row counts, or finding records with no match.
5	WHERE versus HAVING	WHERE filters rows before grouping; HAVING filters groups after aggregation and can use aggregate functions.	Putting SUM() or COUNT() in a WHERE clause.	A condition on a total, count, or average per group.
6	Triggers, functions and stored procedures	A trigger runs automatically when a table event (INSERT, UPDATE, DELETE) occurs; a stored procedure is called explicitly and can run transactions; a function returns a value and can be used in queries.	Expecting a stored procedure to run by itself when data changes.	"Automatically whenever a row is inserted".
7	Slowly changing dimensions	Type 1 overwrites the old value and keeps no history; Type 2 adds a new row with effective dates or a current flag to keep full history.	Overwriting a customer's region when past sales must stay tied to the old region.	Whether history of an attribute must be preserved.
8	Index trade-offs	Indexes speed up reads on columns used in WHERE, JOIN, and ORDER BY, but slow inserts and updates and use storage; they help most on selective columns.	Indexing every column, or indexing a column with two values in a write-heavy table.	A slow query filtering on one column, or a slow bulk load.

Course scope (public description): Advanced Data Management enables learners to extract and analyze raw data. Skillful data management allows organizations to discover and explore data in ways that uncover trends, issues, and their root causes. In turn, businesses are better equipped to capitalize on opportunities and more accurately plan for the future. As organizations continue to extract larger and more detailed volumes of data, the need is rapidly growing for IT professionals who possess data management skills. The skills gained in this course include performing advanced relational data modeling as well as designing data marts, lakes, and warehouses. This course will empower learners with the skills to build business logic at the database layer to employ more stability and higher data-processing speeds. Learners will gain the ability to automate common tasks to summarize and integrate data as they prepare it for analysis. Data Management - Foundations is a prerequisite for this course.